

Python Pit - Coding Challenge

Event structure (5h)

Time	Activity
15:00–15:30	Opening, Rules, Team assembly (if needed), briefing.
15:30–17:00	Development time #1 (Read content, planning, programming)
17:00–17:30	Break (snacks, socializing)
17:30–19:00	Development time #2 (Programming, bug testing, enhancements)
19:00–19:10	Final submissions and tourney setup
19:10–20:00	Tournament, podium and prize ceremony

Context:

You'll write a bot that plays a simple grid-based game (Snake/Tron style arena) against bots written by other students. The goal is to practice turning a fixed set of rules into a working bot, dive deeper into pathfinding algorithms and test your code against your peers in a single elimination tournament.

Objective:

Write one Python file implementing a bot that survives as long as possible against opposing bots, following the rules and constraints of the challenge.

Game rules:

Read the **README.md** file!

Requirements:

- Read the **README.md** file!
- A single .py file named bot_<yourname>.py, copied from bots/template_bot.py.
- It must define BOT_NAME, BOT_COLOR and a get_move(state) function returning "UP", "DOWN", "LEFT", or "RIGHT".
- The exact content of 'state' is present in the **README.md**.

Constraints:

Read the **README.md** file!

- Forbidden to use any LLM's or any type of AI assistance with the exception of CS50's duckAI (<https://cs50.ai/>), the goal is to practice in a challenging environment.

Tournament Rules:

- Single elimination bracket.
- Best out of 3.
- Side switch on each round.
- Retry battle if it's a tie. (After 3 ties, a coin-flip decides the match)